

Homework

Modern Physics

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(Dated: February 19, 2026)

For exercises from the textbook, the corresponding number is reported. The text may have been edited for clarity.

Some of the following exercises (possibly with different numbers) may be on the **mini-test** taking place on the Monday of the following week, or on the **midterms**.

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WEEK I

THE EXPERIMENTAL BASIS OF QUANTUM PHYSICS & STRUCTURE OF THE ATOM

1. Line spectra / Problem 3.11

A transmission diffraction grating with 420 lines/mm is used to study the light intensity of different orders (n). A screen is located $\ell = 2.8$ m from the grating. What are the positions on the screen of the first three red lines corresponding to the $\lambda = 656.2$ nm spectral line of a hydrogen source?

2. Satellite thermal management

A satellite has a flat solar panel with area $A = 20$ m², compared to which we can neglect the surface area of the rest of the satellite.

- Assuming that the satellite is floating in deep space (no solar radiation, approximately 0 K environment), calculate the internal heating power needed to keep it at $T = 270$ K temperature.
[Hint: Don't forget that the solar panel has two sides, assume that it is a perfect black body, use the Stefan-Boltzmann law with $\sigma = 5.67 \times 10^{-8}$ Wm⁻²K⁻⁴.]
- The satellite is in low Earth orbit over the night side of the planet. The solar panel is horizontal, one side is facing Earth, which has a temperature $T_E = 300$ K, the other is facing the night sky at approximately 0 K. Calculate the equilibrium temperature of the satellite with no internal heating.
[Hint: Cancel all the parameters you can, the answer hardly depends on any of the details.]
- The satellite is in low Earth orbit over the day side of the planet, where the sun is directly overhead. One side of the solar panel is facing Earth, which has a temperature $T_E = 300$ K, the other is facing the Sun, receiving radiative power density of $R_{Sun} = 1400$ W/m². Calculate the equilibrium temperature of the satellite with no internal heating.

3. Lightbulb / Problem 3.23

A tungsten filament of a typical incandescent light-bulb operates at a temperature near 3000 K.

- At what wavelength is the intensity at its maximum?
- What is the surface area of the filament, if the light-bulb emits a total of 100 W of radiative power?

4. FM radio / Problem 3.32

An FM radio station of frequency 98.1 MHz puts out a signal of 50.000 W. How many photons/s are emitted?

5. Photoelectric effect / Problem 3.34

What is the threshold frequency for the photoelectric effect on lithium ($\phi = 2.93$ eV)? What is the stopping potential if the wavelength of the incident light is 380 nm?

6. X-ray machine / Problem 3.43

What is the minimum x-ray wavelength produced for a dental x-ray machine where the cathode-ray device is operated at 30 kV?

7. Rutherford Scattering / Problem 4.5

Calculate the impact parameter for scattering a 7.7 MeV α particle from gold at an angle of (a) 1° and (b) 90°.

8. X-ray Scattering / Problem 5.1

X-rays scattered from a crystal have a first-order diffraction peak at 12.5°. At what angle will the second- and third-order peaks corresponding to the same crystal plane appear?